

Model Evaluation

- Confusion Matrix
 - Table
 - Actual vs Predicted
- Metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Support

classification
report

Confusion matrix - bu classification modelning natijalarini baholash uchun kiritiladigan jadval. U modelning to'g'ri va noto'g'ri natijalarini baholash yilganini ko'rsatadi.

y-actual	y-pred	sample
1	1	1
0	0	2
1	0	3
0	0	4
pl	1	5

0 - Negative
1 - Positive

Caption

	Predicted 0	Predicted 1
Actual 0	True Negative	False Positive
Actual 1	False Negative	True Positive

	Predicted 0	Predicted 1
Actual 0	2	0
Actual 1	1	2

from sklearn.metrics import confusion_matrix, classification_report

y_actual = [1, 0, 1, 0, 1]

y_pred = [1, 0, 0, 0, 1]

cm = confusion_matrix(y_actual, y_pred)

array([[2, 0],
[1, 2]])

Accuracy - modelning nechta baho-
ratni to'g'ri qiymatlarini ko'rsatishi-
ni o'ldiray baxolash metrikasi.

$$Acc = \frac{True\ Positive + True\ Negative}{True\ Pos + True\ Neg + False\ Pos + False\ Neg}$$

$$Acc = \frac{2+2}{2+2+0+1} = \frac{4}{5} = 0.8 \cdot 100\% = 80\%$$

Caption

Precision - model Positive deb por-
tiot qilgan natijalarining qanchasi
haqiqatdan ham Positive ekanligini
ko'rsatadi.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

$$\text{Precision} = \frac{2}{2+0} = 100\% = 100\%$$

Recall - model haqiqiy Positive
holatlarining qanchasini topdi
gacha ko'rib bardi.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Neg.}}$$

$$\text{Recall} = \frac{2}{2+1} = \frac{2}{3} = 0.67 = 67\%$$

Precision va Recall farqi

Precision: Model Positive deganlarining
qanchasi to'g'ri?

Recall: Haqiqiy Positive larining qan-
chasi model topdi?

Recall - modelning haqiqiy Positive
holatlarini topish qobiliyatini ko'rsatadi.

Caption

F1-Score. — bu Precision va Recall ning muvozanatli (harmonik o'rtacha) ko'rsatkichi.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1 = 2 \cdot \frac{1 \cdot 0.67}{1 + 0.67} = 0.86 \rightarrow 86\%$$

Classification Report. — bu classification modelning ishlashini baholovchi hisobot bo'lib, unda Precision, Recall, F1-score va Support ko'rsatkichlari har bir sinf uchun alohida beriladi.

Logistic Regression:

Logistic Regression — sigmoid function / Reg

$$\sigma = \frac{1}{1 + e^{-z}} \quad e = 2.71$$

$$z = d + d_0 \cdot x_0 + d_1 \cdot x_1 + \dots + d_n \cdot x_n$$

Caption